

Introduction

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Distributed System

- A **distributed system** is a collection of independent computers (nodes) that **work together and appear to users as a single system**.
- The computers communicate over a network and coordinate their actions **to achieve a common goal**.
- Example:
- When using a service like **Google Search**, your request is processed by many servers located in different data centers. To the user, it looks like one system, but internally **multiple machines collaborate to provide the result**.
- Other examples include:
 - Amazon Web Services (AWS) cloud services
 - Netflix streaming platform
 - Facebook social network
 - Distributed databases such as Apache Cassandra and MongoDB

Characteristics of Distributed System

1. Resource Sharing

Multiple computers share resources such as:

- Files
- Databases
- Printers
- Processing power

2. Concurrency

Many users and processes can access the system simultaneously.

3. Scalability

The system can grow by adding more nodes.

Characteristics of Distributed System

4. Fault Tolerance

If one node fails, other nodes can continue providing service.

5. Transparency

Users see a single system even though multiple machines are involved.

Types of Distributed Systems

1. Client-Server Systems

- Clients request services from servers.
- Example:
 - Web browsers communicating with web servers.

2. Peer-to-Peer (P2P) Systems

- Each node can act as both client and server.
- Example:
 - BitTorrent

Types of Distributed Systems

3. Distributed Computing Systems

- Multiple computers work together to solve computational tasks.
- Example:
 - Scientific simulations
 - Cloud computing platforms

4. Distributed Storage Systems

- Data is stored across multiple machines.
- Examples:
 - Hadoop Distributed File System (HDFS)
 - Google File System (GFS)

Advantages and Disadvantages

Advantages

- ✓ High availability
- ✓ Scalability
- ✓ Resource sharing
- ✓ Better performance through parallel processing
- ✓ Fault tolerance

• Disadvantages

- ✗ Complex design and management
- ✗ Network dependency
- ✗ Security challenges
- ✗ Difficult debugging and synchronization
- ✗ Data consistency issues

Challenges in Distributed Systems

Consistency

- Keeping data synchronized across replicas.

Availability

- Ensuring services remain accessible.

Partition Tolerance

- Handling network failures between nodes.

Clock Synchronization

- Different machines may have different system times.

Fault Handling

- Detecting and recovering from node failures.

Real World Examples

System	Purpose
Google	Search engine
Netflix	Video streaming
Amazon	E-commerce
Dropbox	Cloud storage
Uber	Ride-sharing platform

Thank You